

Errata RSM pg. 45

$$\begin{aligned}
 X_{\text{add}} &= \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}; & X_{\text{int}} &= \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}; \\
 & \begin{pmatrix} 40.444 \\ 0.775 \\ 0.325 \end{pmatrix} & & \begin{pmatrix} 40.444 \\ 0.775 \\ 0.325 \\ -0.025 \end{pmatrix}
 \end{aligned}$$

Remark: The estimates of the additive terms do not change if we add the interaction term. The Design is Orthogonal